

Engineering Mathematics III

ENGE702

Lecture 13: Systems of Differential Equations

Systems of Differential Equations

In this lecture we will look at **systems of linear differential equations** with constant coefficients. In this course we will only consider systems with two equations. They will be of the form:

$$y_1' = a_{11}y_1 + a_{12}y_2$$

$$y_2' = a_{21}y_1 + a_{22}y_2$$

Note that this can be written in the form:

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}' = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

which we often abbreviate as

$$\mathbf{y}' = A\mathbf{y}$$

Systems of Differential Equations

Solving this system means finding functions y_1 and y_2 that satisfy the system of equations.

We have a system of differential equations

$$\mathbf{y}' = A\mathbf{y}.$$

We are dealing with vectors and matrices now, but this looks similar to the differential equation

$$y' = ky$$

for which the solution was

$$y = Ce^{kt}.$$

We will assume it is something like this, and see if we can make it work.

Systems of Differential Equations

We will assume a solution to the system is of the form

$$\mathbf{y} = \mathbf{x}e^{\lambda t}$$

for some vector $\mathbf{x}$ and some constant λ . Then

$$\mathbf{y}' = \lambda \mathbf{x}e^{\lambda t},$$

but we also know from the formulation of our problem that $\mathbf{y}' = A\mathbf{y}$, so we put that together with $\mathbf{y} = \mathbf{x}e^{\lambda t}$, and we have

$$\mathbf{y}' = A\mathbf{x}e^{\lambda t}$$

So

$$A\mathbf{x}e^{\lambda t} = \lambda \mathbf{x}e^{\lambda t}$$

so we need a constant λ and a vector $\mathbf{x}$ such that

$$A\mathbf{x} = \lambda \mathbf{x}.$$

Systems of Differential Equations

This is exactly finding an eigenvalue and eigenvector of A ! Note for two equations our general solution will have two solutions, these come from the two eigenvalues and associated eigenvectors.

Our general solution to our system of equations is then

$$\mathbf{y} = c_1 \mathbf{x}_1 e^{\lambda_1 t} + c_2 \mathbf{x}_2 e^{\lambda_2 t}$$

where $\mathbf{x}_1 = \begin{pmatrix} x_{11} \\ x_{21} \end{pmatrix}$ and $\mathbf{x}_2 = \begin{pmatrix} x_{12} \\ x_{22} \end{pmatrix}$ are eigenvectors of matrix A corresponding to eigenvalues λ_1 and λ_2 respectively, and c_1 and c_2 are arbitrary constants (which we could find if we had initial conditions).

This vector equation can then read:

$$y_1 = c_1 x_{11} e^{\lambda_1 t} + c_2 x_{12} e^{\lambda_2 t}$$

$$y_2 = c_1 x_{21} e^{\lambda_1 t} + c_2 x_{22} e^{\lambda_2 t}$$

Systems of Differential Equations

Example: Find the general solution to the following system of differential equations:

$$y_1' = 7y_1 - y_2$$

$$y_2' = 3y_1 + 3y_2$$

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$$y_1' = 7y_1 - y_2$$

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We can express this as

$$\mathbf{y}' = \begin{pmatrix} 7 & -1 \\ 3 & 3 \end{pmatrix} \mathbf{y}.$$

Now we need to find the eigenvalues and eigenvectors of the matrix $\begin{pmatrix} 7 & -1 \\ 3 & 3 \end{pmatrix}$. Start by solving $\det(A - \lambda I) = 0$. We have

$$(7 - \lambda)(3 - \lambda) + 3 = 0$$

$$\lambda^2 - 10\lambda + 24 = 0$$

$$\lambda = 6 \quad \text{or} \quad \lambda = 4$$

Systems of Differential Equations

Now we can find the eigenvectors:

For $\lambda = 6$:

$$\begin{pmatrix} 1 & -1 \\ 3 & -3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

If $x_1 = 1$ then $x_2 = 1$ will work. So $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is our first eigenvector.

For $\lambda = 4$:

$$\begin{pmatrix} 3 & -1 \\ 3 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

If $x_1 = 1$ then $x_2 = 3$ will work. So $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$ is our second eigenvector (for $\lambda_2 = 4$).

Systems of Differential Equations

Our general solution will then be:

$$\mathbf{y} = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{6t} + c_2 \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{4t}$$

Systems of Differential Equations

Example: Solve the initial value problem:

$$y_1' = 7y_1 - y_2$$

$$y_2' = 3y_1 + 3y_2$$

where $y_1(0) = 1$, $y_2(0) = 0$.

Systems of Differential Equations

Example: Solve the initial value problem:

$$\begin{aligned}y_1' &= 7y_1 - y_2 \\ y_2' &= 3y_1 + 3y_2\end{aligned}$$

where $y_1(0) = 1$, $y_2(0) = 0$.

As usual, to solve an initial value problem we would need to first find the general solution, and then apply the initial conditions to get values for the constants.

In this case the system is the same as the last one, so we can just use the general solution we just found:

$$\mathbf{y} = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{6t} + c_2 \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{4t}.$$

We know that $y_1(0) = 1$ and $y_2(0) = 0$, so applying this we get:

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{(6)(0)} + c_2 \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{(4)(0)}.$$

Systems of Differential Equations

This gives us

$$1 = c_1 + c_2$$

$$0 = c_1 + 3c_2$$

Solving this system of equations gives $c_1 = \frac{3}{2}$ and $c_2 = -\frac{1}{2}$, so our particular solution becomes:

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \frac{3}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{6t} - \frac{1}{2} \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{4t}.$$

Systems of Differential Equations

Exercise: Solve the following initial value problem:

$$\begin{aligned}y_1' &= y_1 - y_2 \\ y_2' &= 2y_1 - 2y_2,\end{aligned}$$

where $y_1(0) = 1$, $y_2(0) = 0$.

Text Book References

Chapters 4.0, 4.1